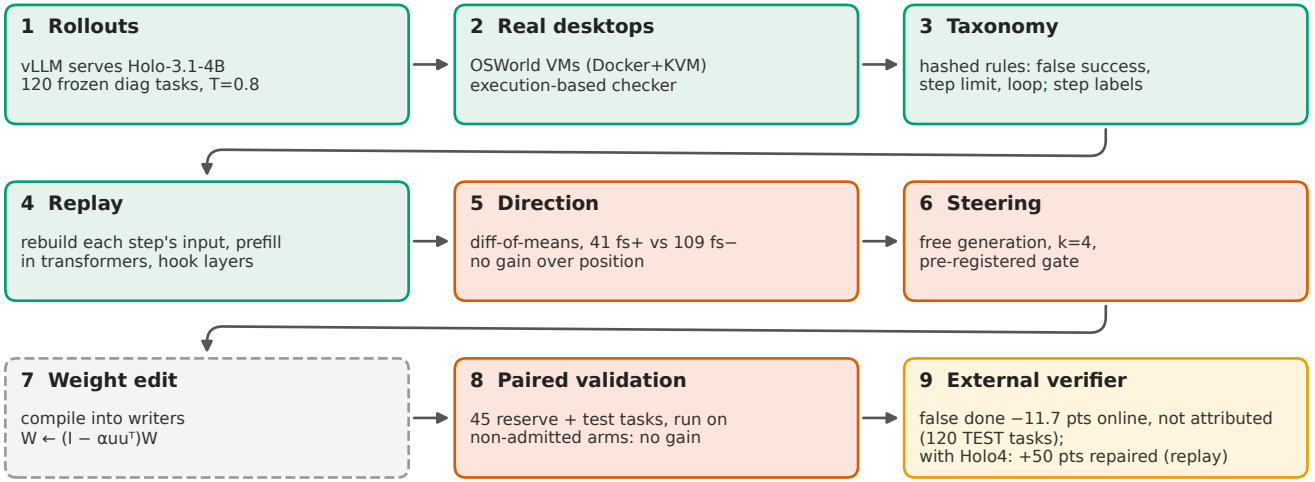
1 Rollouts
vLLM serves Holo-3.1-4B
120 frozen diag tasks, T=0.8
2 Real desktops
OSWorld VMs (Docker+KVM)
execution-based checker
3 Taxonomy
hashed rules: false success,
step limit, loop; step labels
4 Replay
rebuild each step's input, prefill
in transformers, hook layers
5 Direction
diff-of-means, 41 fs+ vs 109 fs−
no gain over position
6 Steering
free generation, k=4,
pre-registered gate
7 Weight edit
compile into writers
W ← (I − αuuᵀ)W
8 Paired validation
45 reserve + test tasks, run on
non-admitted arms: no gain
9 External verifier
false done −11.7 pts online, not attributed
(120 TEST tasks);
with Holo4: +50 pts repaired (replay)
done
run: negative, or no selective effect
ready, not reached
run: offline positive, online not attributed